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Automated Opal multiplex immunofluorescence and PhenoCode Signature Panel staining on the Parhelia Spatial Station

enables reproducible, high-quality spatial phenotyping of the tumor microenvironment in FFPE tissues

Abstract

Multiplex immunofluorescence (mIF) using Quanterix (formerly Akoya) Biosciences Opal tyramide signal amplification (TSA) chemistry and PhenoCode Signature Panels (PSP) provides detailed spatial characterization of the tumor microenvironment (TME) in formalin-fixed paraffin-embedded (FFPE) tissues. However, manual execution of these multi-step staining protocols is labor-intensive, time-consuming, and susceptible to inter-operator variability. Here, we demonstrate the successful automation of both custom Opal mIF panels and pre-designed PhenoCode Signature Panels on the Parhelia Spatial Station. Using human lung cancer and tonsil FFPE specimens, we show that automated staining produces high-quality, spectrally resolved, multiplex fluorescence images with distinct signal for each Opal channel. The results demonstrate clear visualization of immune and tumor markers across 7–8-plex panels, validating the Parhelia Spatial Station as a robust platform for automated Opal and PSP workflows that improves reproducibility while reducing hands-on time.

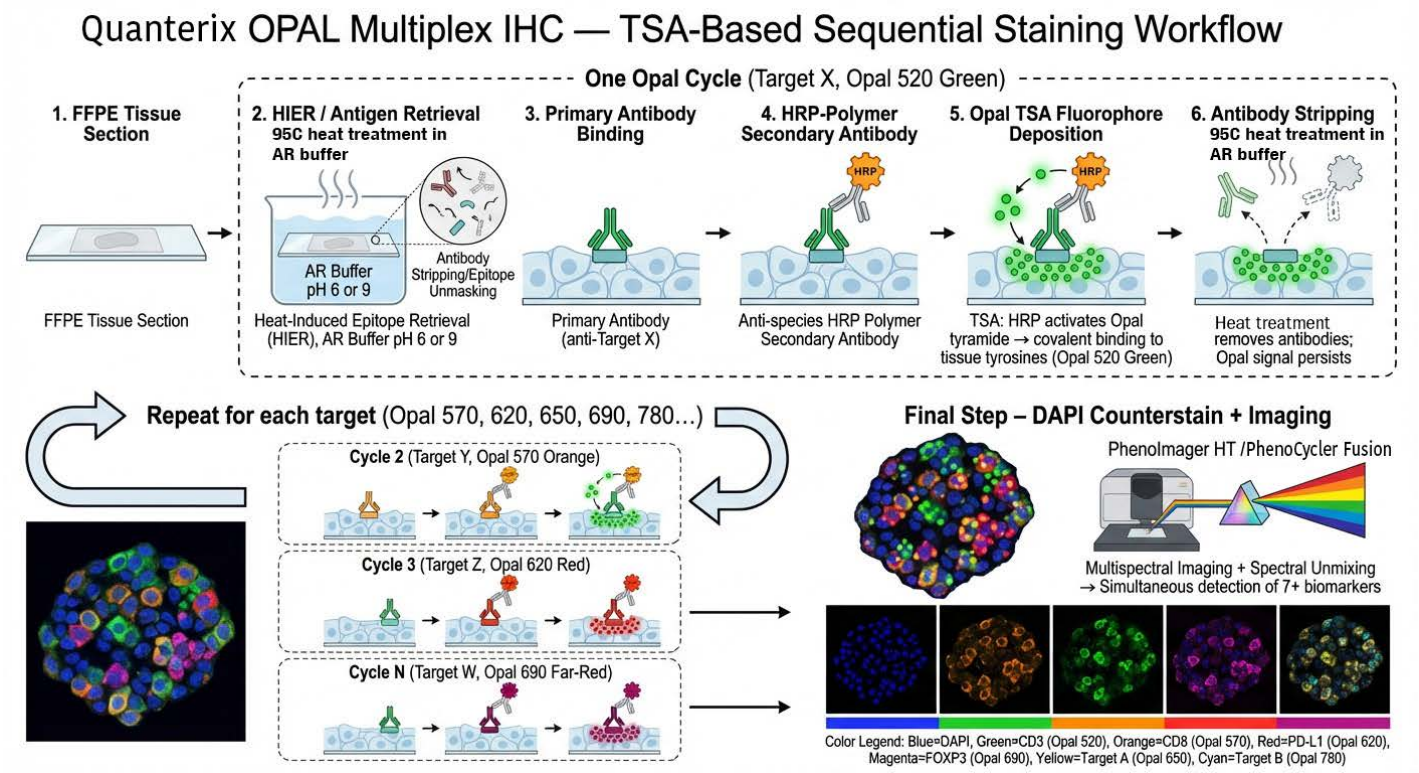

Figure 1. *Schematic diagram of Quanterix (formerly Akoya) OPAL staining technology.*

Introduction

Spatial biology has rapidly advanced our ability to characterize the cellular composition and architecture of the tumor microenvironment (TME). Multiplex immunofluorescence (mIF) using tyramide signal amplification (TSA) enables the simultaneous detection of multiple biomarkers within a single FFPE tissue section, preserving crucial spatial context that is lost with techniques requiring tissue dissociation.

Quanterix (formerly Akoya) Biosciences' Opal TSA chemistry is a well-established platform for mIF, supporting the covalent deposition of spectrally distinct fluorophores at protein target sites (**Figure 1**). This approach enables the construction of multiplex panels of up to 8+ markers (including DAPI nuclear counterstain) while maintaining excellent signal-to-noise ratios and minimal spectral crosstalk. The more recent PhenoCode Signature Panels (PSP) combine barcode-based antibody chemistry from the PhenoCycler platform with Opal TSA signal amplification, providing pre-validated 6-plex kits with one customizable open channel.

Despite these technological advances, manual execution of Opal and PSP staining protocols remains a significant bottleneck. The dewaxing, antigen retrieval followed by iterative cycles of blocking, primary antibody incubation, secondary antibody incubation Opal dye deposition and primary antibody complex stripping (done by microwave or heat treatment) are both time-consuming (often exceeding 12 hours of active hands-on time for a single panel) and prone to variability between operators and staining sessions.

The Parhelia Spatial Station is a purpose-built, open-chemistry robotic platform designed to automate multi-step spatial biology sample preparation workflows. Its capillary gap staining technology ensures uniform reagent coverage while minimizing consumption, and its precision temperature control (0–100°C) supports onboard dewaxing, heat-induced epitope retrieval (HIER), heat driven antibody stripping and temperature-sensitive incubation steps. Here, we report the application of the Parhelia Spatial Station to automate both custom Opal mIF panels and PhenoCode Signature Panel staining across multiple tissue types.

Materials and Methods

Tissue specimens

Human lung cancer FFPE sections were provided by the City of Hope Comprehensive Cancer Center (COH) and Wayne State University. Human tonsil FFPE sections were prepared at Wayne State University. All tissues were sectioned at 4 µm and mounted on positively charged glass slides.

Custom Opal panel design

Three custom Opal panels were designed and optimized. Panel 1 (COH, lung cancer) comprised a 7-plex configuration targeting: usp22 (Opal 480), CD86 (Opal 520), usp7 (Opal 570), CD206 (Opal 620), aquaporin (Opal 690), and sftpc (Opal 780), plus DAPI nuclear counterstain. Panel 2 (Wayne State, lung cancer) targeted: T-bet (Opal 480), RORgt (Opal 520), CD25 (Opal 570), GATA3 (Opal 620), Foxp3 (Opal 690), and CD4 (Opal 780), with DAPI and autofluorescence (AF) channels. Antibody-Opal pairing was guided by target abundance and fluorophore brightness (see Table 1).

PhenoCode Signature Panel

The pan-immune PhenoCode Signature Panel was configured as a modified PSP 6-plex on tonsil tissue with the following targets: CD8 (Opal 480), Ki67 (Opal 520), PD-1 (Opal 570), CD20 (Opal 620), Foxp3 (Opal 690), and CD4 via TSA-DIG amplification with Opal 780 readout. Antibody dilutions were optimized per the manufacturer's recommendations (see Table 2).

Automated staining on the Parhelia Spatial Station

All staining was performed on the Parhelia Spatial Station using the ST12 Slide Staining Module with capillary gap exchange. Protocols were designed and exported via Parhelia StainWorks software. The automated workflow included: (1) onboard dewaxing using Parhelia Dewax reagent; (2) heat-induced epitope retrieval (HIER) at 95°C in AR6 or AR9 buffer; (3) sequential cycles of blocking, primary antibody incubation, HRP-conjugated secondary antibody incubation, and Opal fluorophore deposition; (4) microwave-assisted antibody stripping between cycles; and (5) DAPI nuclear counterstaining and coverslipping.

Imaging

Stained slides were imaged on an Quanterix (formerly – Akoya) PhenoImager system. Whole-slide multispectral images were acquired and spectrally unmixed to resolve individual Opal channels. Composite images were generated for visualization of marker co-expression and spatial distribution patterns.

Results

Parhelia ST12 Slide Staining Module: Precision Thermal Engineering for Multiplexed IHC

Key to the performance of the Parhelia Spatial station is the Parhelia ST12 Slide Staining Module (**Figure 2**) — an engineering solution purpose-built for the demands of complex, multistep staining protocols. The module uniquely combines prolonged low-temperature incubations, which promote the uniform distribution of staining reagents and eliminate concentration gradients across the tissue section, with repeated high-temperature stripping cycles reaching up to 98°C. This broad

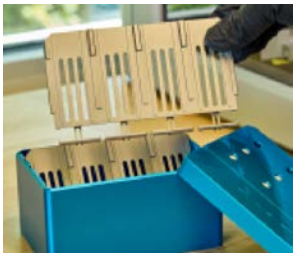

Figure 2. *Parhelia ST12 Slide Staining Module.*

thermal range is made possible by a fully sealed, internally humidity-controlled chamber that maintains optimal moisture throughout the run, paired with precision-engineered slide support shelves that ensure homogeneous heat distribution across every slide position. Together, these design principles address the critical technical requirements of OPAL multiplexed immunofluorescence — making the Parhelia the ideal platform for the reliable, reproducible, and automated execution of OPAL staining workflows on the Spatial Station.

Custom Opal mIF on human lung cancer (COH)

The automated 7-plex Opal staining of human lung cancer tissue produced a richly detailed composite image revealing the heterogeneous spatial distribution of immune and tumor-associated markers (**Figure 3**). All seven Opal channels demonstrated strong, specific signal with clear spectral separation. The composite image reveals distinct tissue compartments: regions of epithelial tumor cells expressing sftpc (Opal 780, shown in magenta/white), interspersed with CD86-positive myeloid cells (Opal 520, green) and CD206-positive macrophages (Opal 620). DAPI nuclear staining provided clear delineation of individual cells throughout the section.

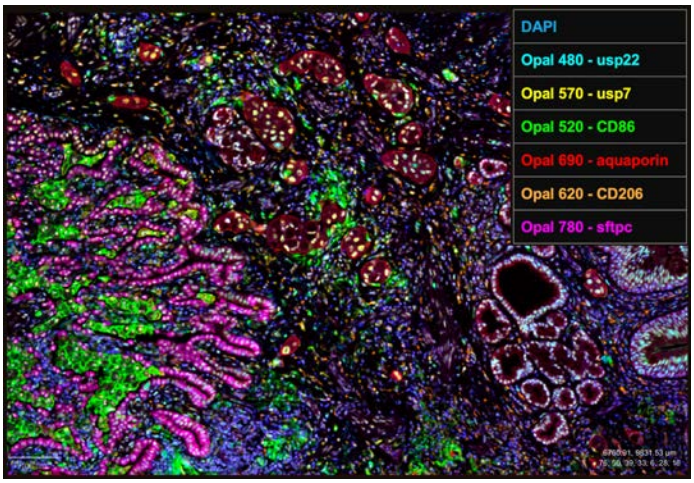

Figure 3. *Automated 7-plex Opal mIF staining of human lung cancer FFPE tissue on the Parhelia Spatial Station. Composite multispectral image showing DAPI (blue), usp22 (Opal 480, cyan), CD86 (Opal 520, green), usp7 (Opal 570, yellow), CD206 (Opal 620, orange), aquaporin (Opal 690, red), and sftpc (Opal 780, magenta). Scale bar: 100 µm. Data provided courtesy of Britney Oeung, City of Hope Comprehensive Cancer Center.*

Custom Opal mIF on human lung cancer (Wayne State)

The Wayne State 7-plex Opal panel targeting T helper cell lineage-defining transcription factors demonstrated successful automated staining of human lung cancer tissue (**Figure 4**). This panel was specifically designed to distinguish CD4+ T cell subsets through their master transcription factors: T-bet (Th1), GATA3 (Th2), RORgt (Th17), and Foxp3 (Treg), alongside CD25 as a T cell activation marker and CD4 for helper T cell identification. The spatial distribution of these markers provides valuable insight into the composition of the adaptive immune infiltrate within the TME.

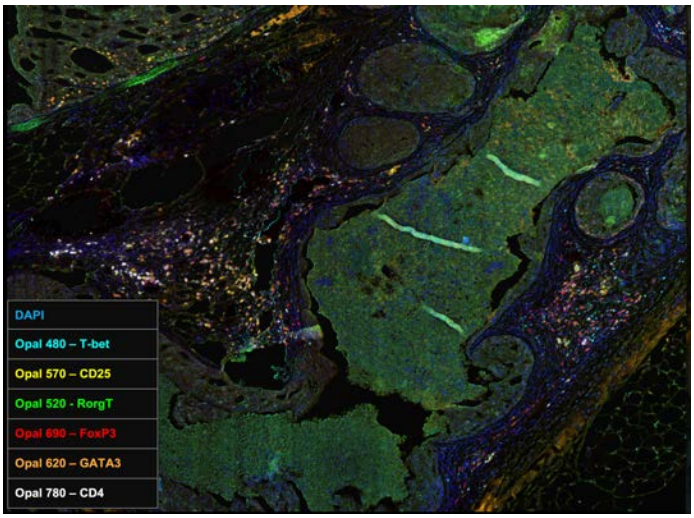

Figure 4. *Automated 7-plex Opal mIF staining of human lung cancer FFPE tissue targeting CD4+ T cell subsets. Composite image showing DAPI (blue), T-bet (Opal 480, cyan), RORgt (Opal 520, green), CD25 (Opal 570, yellow), GATA3 (Opal 620, orange), Foxp3 (Opal 690, red), and CD4 (Opal 780, white), with autofluorescence (AF) subtraction. Data provided courtesy of Jessica Back, PhD and George Schwenkel, Wayne State University, September 2025.*

An additional field of view from the same lung cancer specimen (**Figure 5**) demonstrates the glandular architecture of the tumor with scattered immune infiltrates. The DAPI channel highlights the overall tissue morphology, while individual Opal channels reveal focal T cell clusters in the peritumoral stroma, consistent with immune exclusion patterns frequently observed in non-small cell lung cancer.

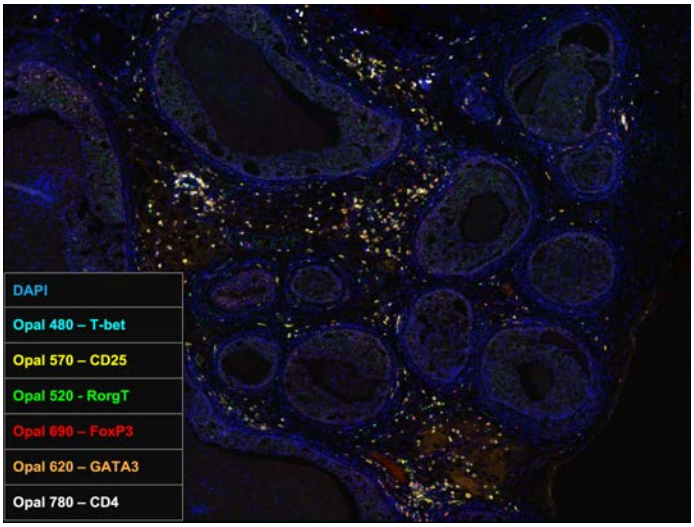

Figure 5. *Additional field of view from automated 7-plex Opal staining of human lung cancer. Same panel configuration as Figure 2, showing glandular tumor architecture with peritumoral immune infiltrates. Data provided courtesy of Jessica Back, PhD and George Schwenkel, Wayne State University.*

PhenoCode Signature Panel on human tonsil

The PhenoCode Signature (PSP) method is different from the regular Opal. It uses primary antibodies conjugated to DNA tags, a reporter oligo with HRP and regular Opal dye, followed by a mild stripping treatment as opposed to high heat.

The automated PhenoCode Signature Panel staining of human tonsil tissue produced a whole-section overview demonstrating uniform staining across the entire specimen (**Figure 6**). The tonsil provides an ideal validation tissue for immune panels due to its well-characterized lymphoid architecture. The composite image reveals the expected spatial organization of immune cell populations: CD20+ B cell follicles, interfollicular T cell zones enriched for CD4+ and CD8+ cells, and scattered Ki67+ proliferating cells concentrated in germinal centers. PD-1 expression is localized to follicular regions consistent with T follicular helper cell distribution, and Foxp3+ regulatory T cells are distributed throughout the T cell zones.

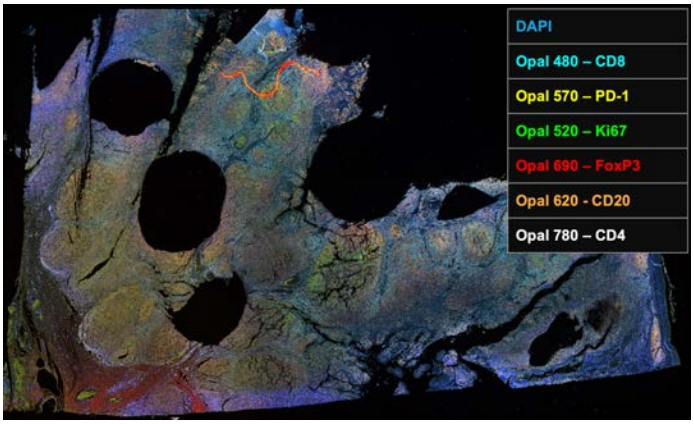

Figure 6. *Automated PhenoCode Signature Panel staining of human tonsil FFPE tissue on the Parhelia Spatial Station. Whole-section multispectral composite showing CD8 (Opal 480, blue), Ki67 (Opal 520, green), PD-1 (Opal 570, yellow), CD20 (Opal 620, orange), Foxp3 (Opal 690, red), CD4 (Opal 780, white), DAPI (blue), and autofluorescence (AF, gray). Scale bar: 600 µm. Data provided courtesy of Jessica Back, PhD and George Schwenkel, Wayne State University, August 2025.*

Panel Configurations

Table 1. *Custom Opal panel configurations used in this study.*

Opal Channel	Panel 1 (COH, Lung)	Panel 2 (Wayne St, Lung)
Opal 480	usp22	T-bet
Opal 520	CD86	RORgt
Opal 570	usp7	CD25
Opal 620	CD206	GATA3
Opal 690	aquaporin	Foxp3
Opal 780	sftpc	CD4
DAPI	Nuclei	Nuclei

Table 2. *PhenoCode Signature Panel configuration (tonsil tissue).*

Target	Dilution	Opal Channel
CD8	1:2000	Opal 480
Ki67	1:100	Opal 520
PD-1	1:75	Opal 570
CD20	1:800	Opal 620
Foxp3	1:100	Opal 690
CD4	1:400	TSA-DIG → Opal 780

Discussion

The results presented here demonstrate that the Parhelia Spatial Station is capable of fully automating both custom Opal multiplex immunofluorescence panels and PhenoCode Signature Panel staining protocols on FFPE tissues. Across all panel configurations and tissue types tested, automated staining produced high-quality multispectral images with strong, specific fluorescent signals and clean spectral separation between channels.

The successful automation of 7–8-plex Opal panels is particularly noteworthy given the complexity of the sequential staining workflow. Each marker requires a complete cycle of blocking, primary antibody incubation, HRP-polymer incubation, Opal dye deposition, and heat-mediated antibody stripping. Manual execution of these steps introduces cumulative variability, especially in temperature control during HIER and stripping steps, incubation timing, and reagent volumes. The Parhelia Spatial Station's capillary gap staining technology ensures uniform reagent coverage with minimal consumption, while its programmable temperature module maintains precise thermal conditions throughout the protocol.

The PhenoCode Signature Panel results on tonsil tissue validate that the barcode-based antibody cocktail approach, which allows all primary antibodies to be applied simultaneously before sequential detection, is fully compatible with automated workflows on the Spatial Station. This is significant because PSP chemistry represents a distinct workflow from traditional sequential Opal staining, requiring different handling of the initial cocktail incubation and subsequent barcode-directed detection steps.

The ability to automate these protocols offers several practical advantages for research laboratories: reduced hands-on time (from approximately 8+ hours to walkaway automation), improved batch-to-batch consistency, precise reagent management through the StainWorks software interface, and the ability to process multiple slides in parallel using the ST12 module. These benefits are especially impactful for large-scale translational studies where staining reproducibility across many patient samples is critical.

Conclusions

We have demonstrated that the Parhelia Spatial Station provides a robust, automated platform for Opal multiplex immunofluorescence and PhenoCode Signature Panel staining of FFPE tissues. Custom 7-plex Opal panels on human lung cancer and a modified 6-plex PSP on human tonsil all produced high-quality multispectral images with clear marker-specific signals. Automation on the Spatial Station substantially reduces manual labor, minimizes operator-dependent variability, and supports standardized workflows suitable for both discovery research and translational studies interrogating the tumor microenvironment.

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